

README.md — Project __NovÆxorpus(NÆX) & Lex-Novi-Æxentis-Copiæ

Master Ecosystem Overview, Federated Multi-Corpora Architecture & Operational Guide

Motto: *Xçineribus, in-variis-nunquam-varius, Novi-Æxentis-Copiæ, Vincent*

(From the ashes, unwavering in the midst of adversity, new forces of abundance conquer)

Living Core: **Æsc & Æyre** — The central body representing the living truth of the system; decoupled native APKs forming the on-device execution foundation alongside Horizons UI.

Universal Memory Engine: **#d.u.m.b.a.s.s.** (Database & Universal Memory Bank Across Split Services)

1. Executive Overview

Project __**NovÆxorpus(NÆX)** and __**Lex-Novi-Æxentis-Copiæ** establish a sovereign, multi-tier cognitive agent architecture spanning edge mobile hardware (Snapdragon 8 Elite Hexagon HTP), dedicated compute servers (NVIDIA Jetson Orin Nano Super), visual terminal workstations (Rubik Pi 3 Dragonwing), and enterprise cloud training pipelines.

This repository serves as the **Federated Master Corpus Root**, organizing codebases, documentation, memory banks, and procedural tool suites into a coherent, self-improving cognitive operating system.

2. Core Taxonomy & Brand Ligatures

- **Horizons UI (horizons-ui):** The primary presentation shell and concierge frontend. Built as an Android Chromium WebView with persistent WebSocket bridges to native daemons.
- **Æsc (aesc) APK:** The native System Terminal & Shell Daemon APK. Serves as a complete replacement for Termux by registering as an OS Accessibility Service and Assistant. Hosts the local **ADB loopback** on **127.0.0.1:5555** to execute shell commands with UID 2000 (**shell**) permissions without root.
- **Æyre (aeyre) APK:** The native Sensory Ingress & Media Daemon APK. Runs bare-metal Silero VAD (Voice Activity Detection), Moonshine Small ONNX STT, Kokoro-82m / Sherpa ONNX TTS, and Video Game SDK screen capture.

- **NovusÆxenti (novus-aexenti)**: The cognitive Multi-Agent MoE brain, coordinating the 0.8B Qwen triage model and 4B/9B query/reasoning models.
- **NovÆxopia (novaecopia)**: "The Claw" — the tool harness runtime, OpenWiki TUI, and MCP capability bridge.
- **Æsop-Xi (aesop-xi)**: Agentic Execution Split Operations Protocol — the tactical orchestration, ethical safety, and behavioral policy governance layer.
- **NovÆxorpus (novae-xorpus)**: The hardened, immutable universal database, LLM wiki vault, and ground-truth repository.
- **#d.u.m.b.a.s.s.: Database & Universal Memory Bank Across Split Services** (*"It isn't 'dumbass-proof' if it hasn't been '#d.u.m.b.a.s.s.' proven"*).

3. The Federated Multi-Corpora Architecture

Instead of forcing an identical file layout onto fundamentally different data types, the ecosystem organizes into a **Federation of Specialized Repositories**:

novae-xorpus/ (FEDERATED MASTER CORPUS ROOT & LIVING TRUTH)

```

|
|
| — 📖 README.md                # Global Ecosystem ReadMe & Bootstrap Guide
|
| — 🗺️ MAP.md                    # Top-level human navigation ontology
|
| — 📊 master_manifest.jsonl     # Federated hash table linking all child repos
|
|
| — 📁 aesop-xi/                 # 1. CORE ORCHESTRATION & ETHICAL REPO
|   | — manifest.jsonl
|   |
|   | — arbitration/             # Resource allocation & priority arbitration logic
|   |
|   | — policies/                # Hard boundaries, permissions, and tool guardrails
|   |
|   | — .incognito_red_sandbox/  # Sealed shadow validation sandbox
|   |
|
| — 📁 novus-aexenti/            # 2. COGNITIVE REASONING & MEMORY REPO

```

— manifest.jsonl	
— dual_agent_router/	# 0.8B Triage vs. 9B Query handshake logic
— mem0_episodic/	# In-session habit keys and rolling preferences
— reasoning_bank/	# System Flywheel (RLVR state tracking & recovery)
— 📁 novaexopia/	# 3. HARNESS RUNTIMES & 3-APK NATIVE REPO
— manifest.jsonl	
— horizons-ui/	# APK 1: Chromium WebView frontend & UI tiles
— aesc/	# APK 2: Terminal daemon, ADB loopback (port 5555)
— aeyre/	# APK 3: Media daemon, Silero VAD, Moonshine, Kokoro
— openwiki-tui-harness/	# Terminal user interface (File Admin / Help Desk)
— modular_harnesses/	# Swarm extensions (Prime Agent RLM, ECC skills)
— mcp_connectors/	# Node.js Filesystem MCP, SQLite MCP, Postgres MCP
— 📁 vendor-corpora/ LIBRARIES)	# 4. VENDOR KNOWLEDGE BASES (IMMUTABLE
— qualcomm-qairt-sdk/ headers	# Qualcomm QAIRT, HTP v79 FastRPC specs &
— google-android-platform/	# Android OS Foreground Services, LMK, Media SDK
— 📁 skills-and-capabilities/	# 5. GENERAL CAPABILITIES & CODE GRAPH REPO
— manifest.jsonl	

- | |— code-review-graph/ # PyGraphify / Graphify AST dependency mappings
- | |— notebook-lmpy/ # Deep corpus analytical query scripts
- | |— obsidian-skills/ # Vault hygiene and graph-sync definitions
- |
- |— 📁 data_vault/ # 6. THE LIVING LLM WIKI & DATA VAULT
 - |— manifest.jsonl
 - |— 01_raw_sources/ # Cold archive (immutable ground truth)
 - |— 02_wiki_md/ # Semantic memory (Zettelkasten conceptual notes)
 - |— 03_recall_cache/ # Working memory accelerator (JSONL & SQLite KV)
 - |— 04_skills_runtime/ # Procedural memory (prompt_skills/ & extracted_tools/)
 - |— 05_episodic_logs/ # Episodic telemetry (trajectories & RLVR verifiers)

4. The 5+1 Unified Cognitive Memory Standard

Tier	Directory	Cognitive Faculty	Operational Role
Root	MAP.md & manifest.jsonl	Metacognition	Master index, global hash registry, cross-tier entity resolution.
Tier 1	01_raw_sources/	Sensory Perceptual Store	Immutable ground truth: whitepapers, manuals, transcripts (pdf/, media/, text/).
Tier 2	02_wiki_md/	Semantic Memory	Human- and AI-readable Markdown graph

Tier	Directory	Cognitive Faculty	Operational Role
			governed via OpenWiki TUI.
Tier 3	03_recall_cache/	Working Memory Accelerator	High-speed machine recall: pre-tokenized JSONL, vectors, and embedded SQLite.
Tier 4	04_skills_runtime/	Procedural Memory	Modular prompt skills (prompt_skills/) and deterministic code tools (extracted_tools/).
Tier 5	05_episodic_logs/	Episodic Telemetry	Trajectory logs, daily driver syncs, RLVR verifiers, and .incognito_red_sandbox/.

5. Dual Operational Modes

- Mode A: Developer Terminal Sessions (Claude Code CLI):**
 - Governing Harness: ECC (Everything Claude Code).**
 - Tool Suite:** honey-crush (large doc ingestion), nexus-mapper (AST code mapping), ecc-planner, and Matt Pocock engineering skills (spec, ticket, tdd, code-review).
 - Prime Agent does not intervene in this mode.
- Mode B: Sovereign Edge Assistant (On-Device Local Weights):**
 - Governing Harness: Prime Agent (from Prime Intellect).**
 - Loop Mechanism:** Recursive Language Model (RLM) running inside a persistent Python REPL.
 - Inference:** Qwen 3.5 models running on Snapdragon 8 Elite Hexagon HTP via Qualcomm GenieX.
 - Continual Harness:** Reset-free online adaptation updating prompt state and memories with automated rollback snapshots.
 - Inter-Mode Bridge:** Prime Agent calls the claude CLI strictly as an external subprocess tool without nested harness wrapping.

6. Master Documentation Index in Google Drive

-  [00_DEFINITIVE_MASTER_SPECIFICATION_V3_COMPLETE.md](#): Full 10-section canonical master system specification.
 -  [01_SOVEREIGN_NODE_AND_APK_TOPOLOGY.md](#): Decoupled 3-APK framework, ADB loopback, and DroidDesk workstation mode.
 -  [02_DUMBASS_UNIVERSAL_MEMORY_SPEC.md](#): `#d.u.m.b.a.s.s.` division of labor (mem0, OB1, OmniRoute, Reasoning Bank, SQLite, Postgres).
 -  [03_DUAL_OPERATIONAL_HARNESS_AND_MCP_SPEC.md](#): Operational modes, MCP runtime servers, and PyGraphify code graphs.
 -  [04_ON_DEVICE_INGESTION_AND_W5H_FRAMEWORK.md](#): W5+H System Matrix, clean Markdown pipeline, and JSONL marker extraction.
 -  [RESUME.md](#): Comprehensive session state and handoff ledger.
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7. Standalone Production Tools Index in Google Drive

-  [system_housekeeper.sh](#): Zero-trust Obsidian, Markor, Drive, and Git hook boundary sweeper.
-  [agent_panel.sh](#): Interactive terminal control panel menu for the `Æsc` daemon.
-  [compile_manifest.py](#): Universal JSONL manifest compiler and SHA256 checksum generator.
-  [doc_to_skill_and_tool.py](#): Automated dual extraction engine converting docs into `SKILL.md` rules and executable tools.
-  [npu_manager.py](#): Snapdragon NPU hot-swap manager over UNIX domain sockets (`/dev/socket/npu_manager.sock`).
-  [export_drive_docs.js](#): Google Apps Script exporting Drive Docs to clean Markdown and JSONL (no PDFs).
-  [log_builder.py](#): Mobile episodic telemetry logger appending to `agent_logs.jsonl`.
-  [convert_raw_to_markdown.py](#): On-device doc normalizer stripping web HTML and adding YAML frontmatter.
-  [generate_jsonl_markers.py](#): Lightweight JSONL marker generator for low-memory RAG index caches.